Pass Scions Handwritten

Setup, forward pass, derivative

$y_n : x = 1$, $y = 2$, $w = 0.5$, $b = 0$, $v = 1$, $c = 0$

$\eta = 0.1$

Model $= z = wx + b$, $h = \tanh(z)$, $\hat{y} = vh + c$, $J = \frac{1}{2}(\hat{y} - y)^2$

Forward Pass

$z = (0.5)(1) + 0 = 0.5$

$h = \tanh(0.5) = 0.462117$

$\hat{y} = 1(0.462117) + 0 = 0.462117$

$J = \frac{1}{2}[0.462117 - 2)^2 = 1.188542$

Back ward Pass

$\partial J / \partial \hat{y} = \hat{y} - y = -1.537883$

$\partial J / \partial c = (\partial J / \partial \hat{y})(\partial \hat{y} / \partial c) = -1.537883(1)$

$= (-1.537883)(0.462117) = -0.710682$

$\partial J / \partial h = = \partial J / \partial \hat{y} \cdot \partial \hat{y} / \partial h = -1.537883(v) = -1.537883$

$\partial h / \partial z = = 1 - 0.213552 = 0.786448$

$\partial J / \partial w = \partial J / \partial z \cdot \partial h / \partial w = (-1.209464)(x = 1) = -1.209464$

$\partial J / \partial b = \partial J / \partial z \cdot \partial z / \partial b = (-1.209464)(1) = -1.209464$

Gradient descent update $\theta_{new} = \theta - \eta \frac{\partial J}{\partial \theta}$

$w = 0.5 - 0.1\ (-1.209464) = 0.5 + 0.120946 = 0.620946$

$b = 0 - 0.1\ (0.209464) = 0.120946$

$V - 1 = 0.1\ (-0.710682) = +0.071068 = 1.071068$

$c = 0 - 0.1\ (-1.537888) = 0.153788$

$Z = 0.620946(1) + 0.12096 = 0.74183$

$A = \tanh(0.74183) = 0.630287$

$J = 1/2\ (0.8288269 - 2)^2 = 0.685774$